Testing and Evaluation (ENG520)

- **Measurement, Assessment and Evaluation**
  - **Measurement**: The process of quantifying student learning or behavior.
  - **Assessment**: The process of collecting and interpreting data to determine student learning or behavior.
  - **Evaluation**: The process of making judgments about student learning or behavior based on assessment data.
- **Classroom Assessment OR Types of Assessment**
  - **Formative**: Ongoing, continuous assessment to monitor student progress.
  - **Summative**: Evaluative assessment to determine student learning at the end of a lesson, unit, or course.
  - **Diagnostic**: Assessment to identify student strengths, weaknesses, and learning needs.
  - **Self-Assessment**: Students evaluate their own learning.
  - **Peer Assessment**: Students evaluate each other's work.
- **Formative and Summative Assessment**
  - **Formative Assessment**: Inform instruction, adjust teaching, and provide feedback to students.
  - **Summative Assessment**: Evaluate student learning, grade, and report progress.
- **Types of Assessment**: Methods of Interpreting Results
  - **Norm-Referenced**: Compare student performance to a larger group.
  - **Criterion-Referenced**: Evaluate student performance based on specific criteria.
  - **Standards-Based**: Evaluate student performance based on learning standards.
- **Role of National Curriculum in Assessment**: Provides a framework for assessment and evaluation. Ensures consistency and standardization.
- **Connecting all Four Levels in Curriculum**
  - **Aims**: Overall goals of education.
  - **Objectives**: Specific learning outcomes.
  - **Learning Outcomes**: Statements of what students will know or be able to do.
  - **Assessment**: Evaluates student achievement of learning outcomes.

- **Modes of Assessment in Curriculum**
  - Written tests and exams.
  - Oral presentations and interviews.
  - Practical and performance tasks.
  - Projects and portfolios.
- **Taxonomies of Educational Objectives and Assessment:**

Bloom‘s Taxonomy:

- Knowledge.
- Comprehension.
- Application.
- Analysis.
- Synthesis.
- Evaluation.

SOLO Taxonomy

- **Surface**: Recall and reproduction.
- **Deep**: Understanding and application.
- **Relational**: Connecting ideas and concepts.
- **Extended**: Critical thinking and problem-solving.

- **Bloom‘s Taxonomy**

A framework for categorizing learning objectives into six levels:

- Remembering (recall knowledge)
- Understanding (comprehend concepts)
- Applying (use knowledge in new situations)
- Analyzing (break down complex information)
- Evaluating (make judgments based on criteria)
- Creating (generate new ideas or products)

- **Revised version of Bloom‘s Taxonomy:**

Updates the original taxonomy to include:

- Remembering
- Understanding
- Applying
- Analyzing
- Evaluating
- Creating

- **Instructional Objectives:** Specific statements that outline what students will learn or achieve. Should be clear, measurable, and aligned with assessments

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Educational Decisions Making

Assessment data informs decisions about:

- Student promotion or retention
- Curriculum development
- Instructional strategies
- Resource allocation

Norm Referenced Assessment (NRT)

- Compares student performance to a larger group (norm)
- Purpose: Rank students, determine relative standing
- Example: Standardized tests like SAT or ACT

Criterion Referenced Assessment (CRT)

- Evaluates student performance based on specific criteria or standards
- Purpose: Determine mastery or achievement of learning objectives
- Example: Quizzes, unit tests, project evaluations

Purpose of Testing

- Evaluate student learning
- Identify strengths and weaknesses
- Inform instruction
- Make educational decisions

Characteristics of Criterion Referenced Assessment

- Focus on specific learning objectives
- Emphasizes mastery or achievement
- Uses clear criteria and standards
- Results are reported in terms of mastery or achievement

Difference Between NRT and CRT

- **NRT**: Compares students to a norm, focuses on relative standing
- **CRT**: Evaluates student performance based on specific criteria, focuses on mastery or achievement

Formative Assessment

- Ongoing, continuous assessment during instruction
- Purpose: Monitor progress, adjust instruction, provide feedback
- Examples: Quizzes, classwork, self-assessment

Functions of Formative Assessment:

- Identify areas of improvement
- Adjust instruction
- Provide feedback to students
- Encourage student self-assessment

Summative Assessment

- Evaluative assessment at the end of a lesson, unit, or course

- Purpose: Determine student learning, grade, or evaluate achievement
- Examples: Unit tests, final exams, project evaluations

Functions of Summative Assessment

- Evaluate student learning
- Determine grades or promotion
- Provide feedback to students and parents
- Evaluate instructional effectiveness

Table of Specification

- A tool for constructing assessments
- Aligns learning objectives with assessment items
- Ensures coverage of all learning objectives

Elements and Appropriateness in Table of Specification

- **Elements**: Learning objectives, assessment items, cognitive levels
- **Appropriateness**: Ensures assessment items align with learning objectives and cognitive levels

Balance Among Learning Objectives and their Weight in Table of Specification with Example

- Ensures that all learning objectives are represented in the assessment
- Weightage is assigned to each objective based on its importance
- **Example**

Learning Objectives

- Knowledge (30%)
- Understanding (20%)
- Application (30%)
- Analysis (20%)

SELECTION OF TEST

Selecting pre-designed standards:

- Use existing standards and frameworks
- Ensure alignment with learning objectives

Selecting appropriate test:

- Consider the purpose of the test
- Choose the right type of test (e.g., multiple-choice, essay)

Fairness in Selecting Appropriate Test

- Ensure the test is fair and unbiased
- Consider the diversity of students and their needs
- Avoid cultural or linguistic biases

Characteristics of Good Test

- **Validity**: Measures what it claims to measure
- **Reliability**: Consistent results over time and conditions
- **Usability**: Easy to administer, score, and interpret

- **Evidences of Validity**
  - **Content validity**: Aligns with learning objectives and content
  - **Construct Validity**: Measures the intended construct or concept
  - **Criterion Validity**: Predicts student performance in future tasks
  - **Consequence Validity**: Has a positive impact on student learning
- **Nature of Reliability**: Consistency of test scores over time and conditions. Depends on the test, test-takers, and testing conditions
- **Method of Estimating Reliability**
  - **Test-retest Method**: Correlates scores from two administrations
  - **Equivalent Form Method**: Correlates scores from two parallel forms
  - **Split Half Method**: Correlates scores from two halves of the test
  - **Kuder-Richardson Method**: Estimates reliability from a single administration
- **Anecdotal Records**: Narrative descriptions of student behavior, performance, or achievements
- **Effective Use of Anecdotal Records:**
  - Use them to supplement other assessment methods
  - Focus on specific behaviors or skills
  - Use them to identify areas of improvement
- **Advantages and Limitations of Anecdotal Records**

Advantages:

- Provide rich, qualitative data
- Offer insights into student behavior and learning
- Can be used to identify strengths and weaknesses

Limitations:

- Subjective and prone to bias
- Time-consuming to collect and analyze
- May not be reliable or generalizable

- **Peer Appraisal**: Students evaluate each other's work or performance
- **Portfolio**: A collection of student work and artifacts that demonstrate learning and progress
- **Purpose of Portfolio**
  - Showcase student learning and progress
  - Encourage student reflection and self-assessment
  - Provide a comprehensive picture of student achievement
- **Selection of Item in a Test**: Choose items that align with learning objectives
- **Characteristics of MCQs**: Clear, concise, and objective

- **Uses of MCQs**: Assess knowledge, comprehension, and application
- **Advantages and Limitations of MCQs**
  - **Advantages**: Efficient, objective, and easy to score
  - **Limitations**: May not measure higher-order thinking, can be vulnerable to guessing
- **Suggestions for Constructing MCQs**
  - Use clear and concise language
  - Avoid ambiguous or complex questions
  - Use plausible distractors
  - Use a variety of question types
- **Uses of True/ False Items**: Assess knowledge and comprehension
- **Advantages and Limitations of True-False**
  - **Advantages**: Easy to construct and score, efficient
  - **Limitations**: May not measure higher-order thinking, can be vulnerable to guessing
- **Suggestions for Constructing True-False (Alternative Form) Items**

- Use clear and concise language
- Avoid ambiguous or complex statements
- Use a variety of question types

Groups by Maha Malik

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